

8(a)(i)	A – label line pointing to mitochondrial matrix ;	1
8(a)(ii)	crista(e) ;	1
8(a)(iii)	<i>any five from:</i> 1 folded (inner) membrane / many cristae ; 2 forms large surface area ; 3 (for) electron carriers / electron transport chain / ETC / cytochromes ; 4 ref. to ATP synth(et)ase / ATP synthesis ; 5 impermeable to, protons / hydrogen ions ; 6 (so) allows, formation of proton gradient / high concentration of protons in intermembrane space ; 7 proton pumps ;	5

Question	Answer	Marks
8(b)	DCPIP / methylene blue ; changes, colour / blue to colourless, when, reduced / accepts electrons (from succinate) ;	2
8(c)(i)	any three from: 1 rate of respiration increases (as succinate concentration increases) up to 1.0 mmol dm ⁻³ or maximum rate at 1.0 mmol dm ⁻³ ; 2 rate of respiration higher at 37 °C than at 10 °C (in liver and muscle) ; 3 rate of respiration higher in muscle than in liver (at both temperatures) ; 4 comparative data quote to support mp2 or mp3 ;	3
8(c)(ii)	any three from: <i>muscles have higher rate of respiration because:</i> 1 respiration produces ATP ; 2 muscle has higher need for, ATP / energy ; ora 3 for muscle contraction or muscle is more metabolically active / AW ; 4 AVP ; e.g. shivering to maintain body temperature ;	3
8(c)(iii)	muscle, is contracted / generates heat / is more metabolically active / AW ; ora	1